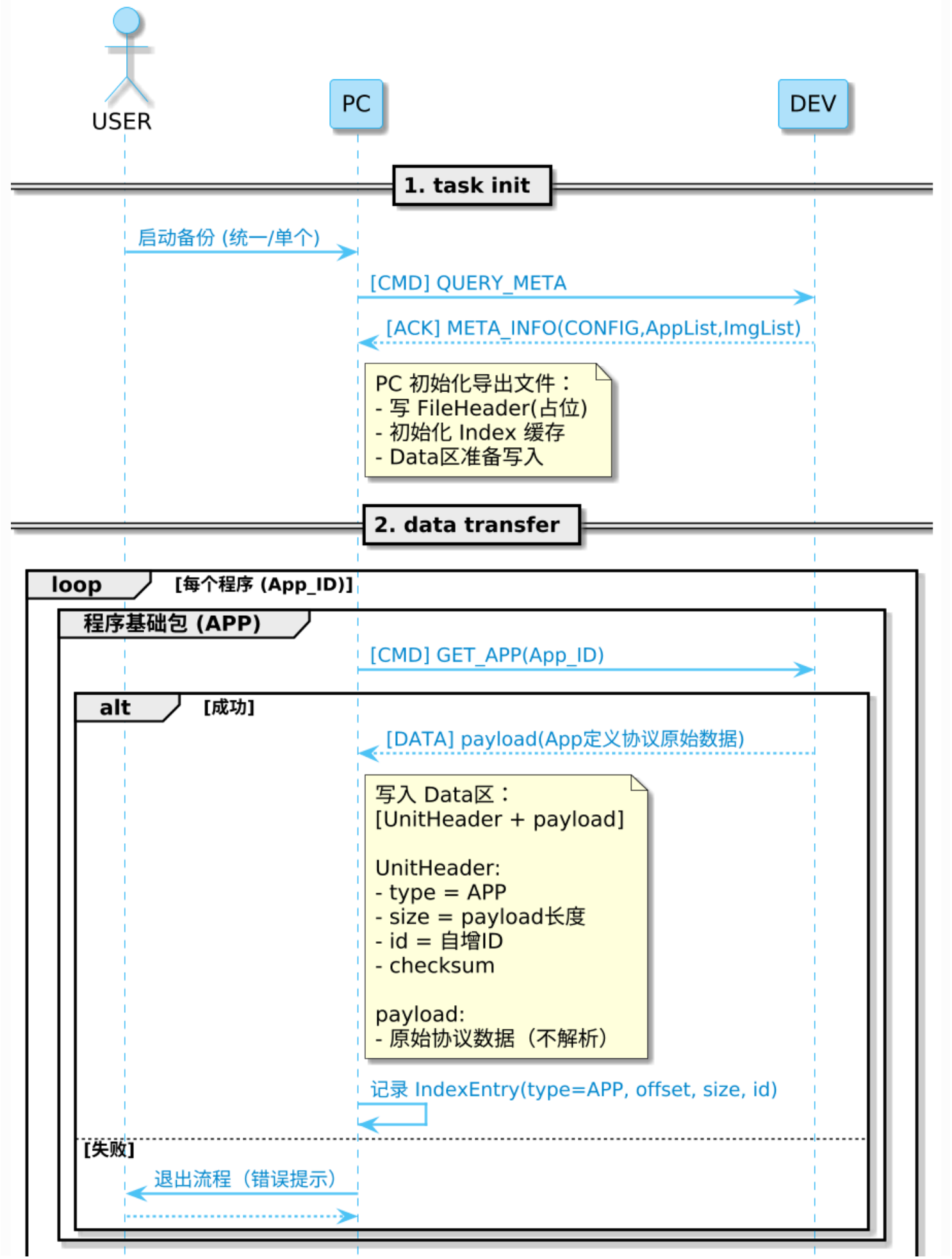
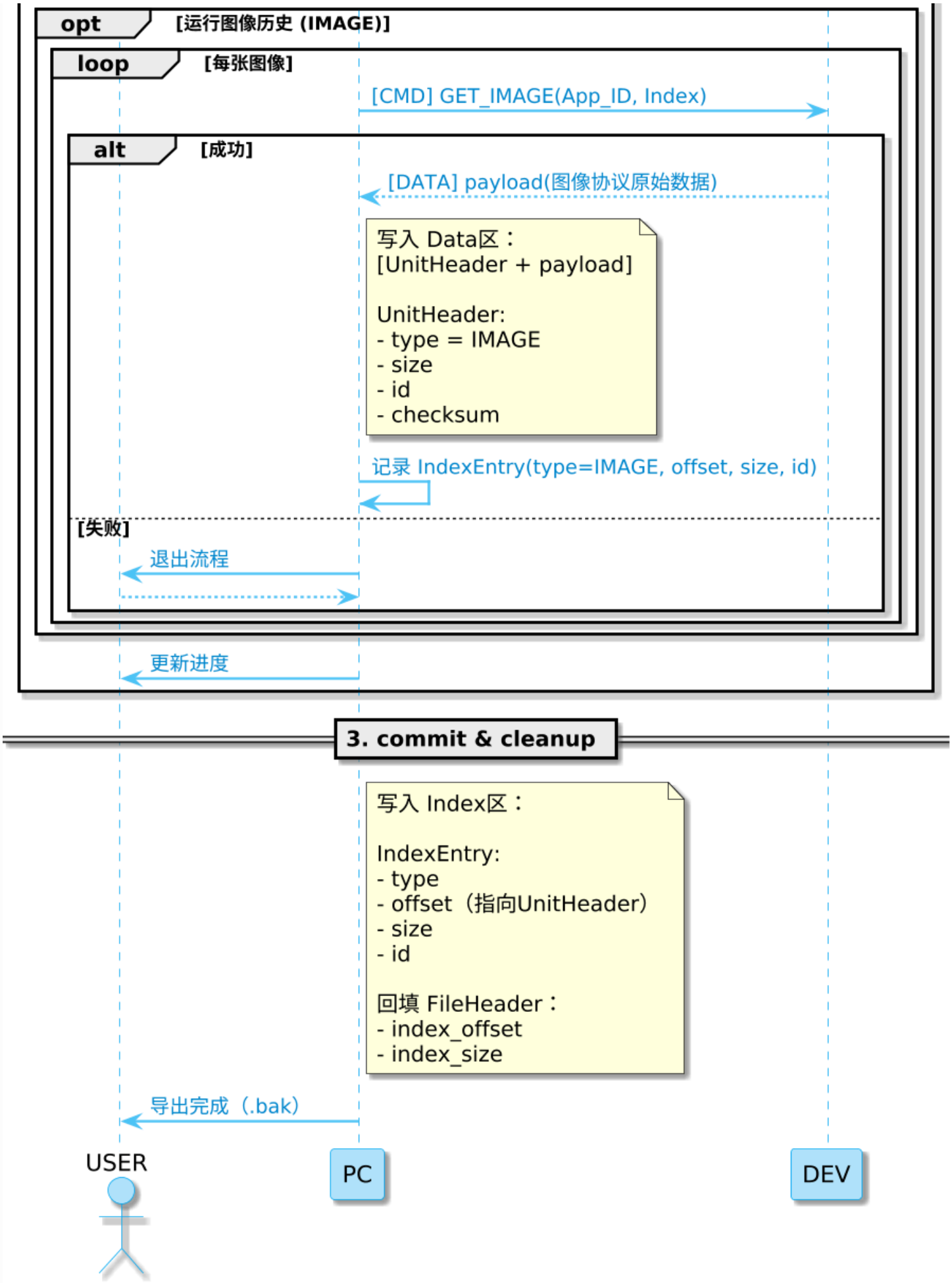
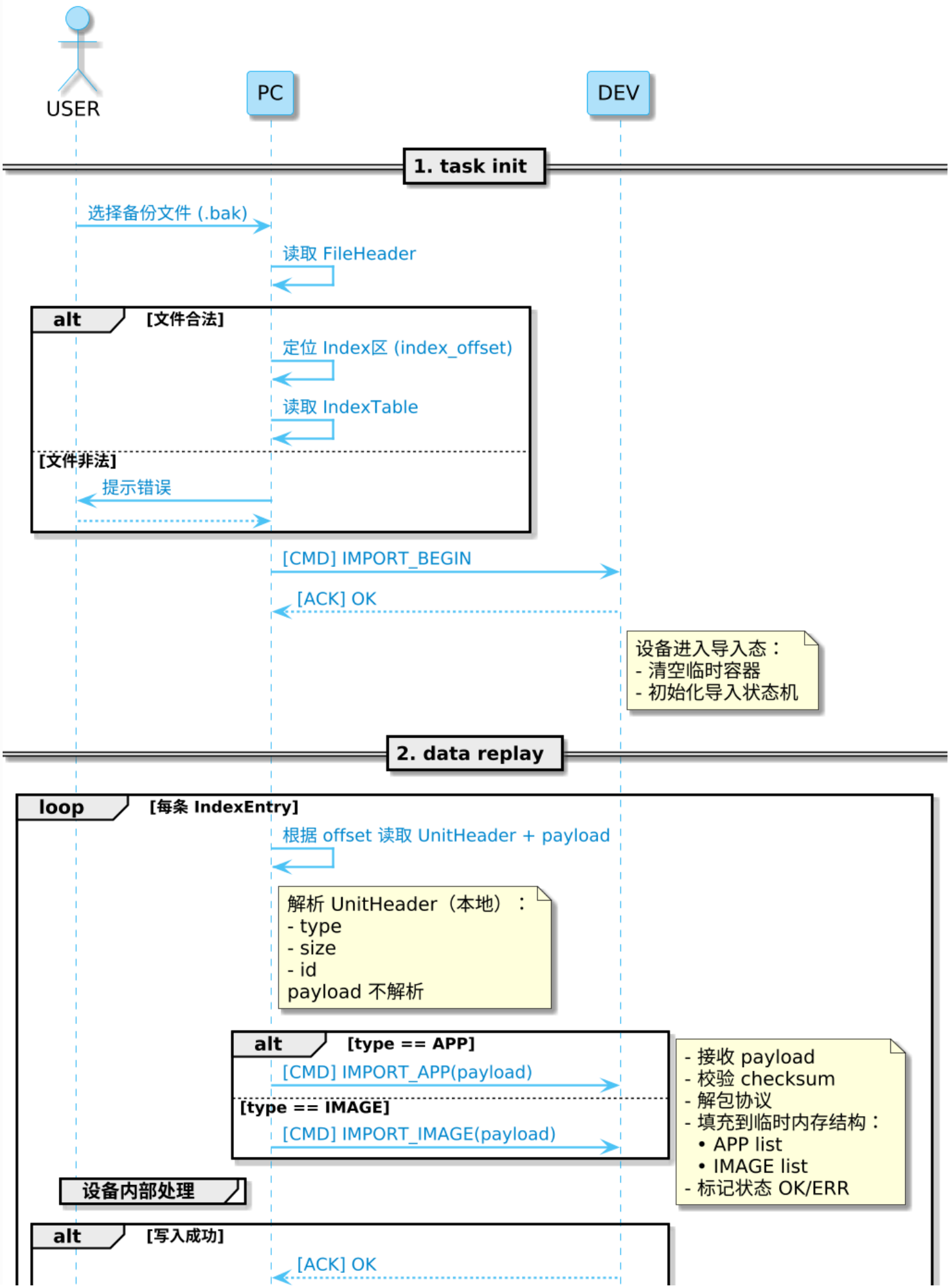
时序图

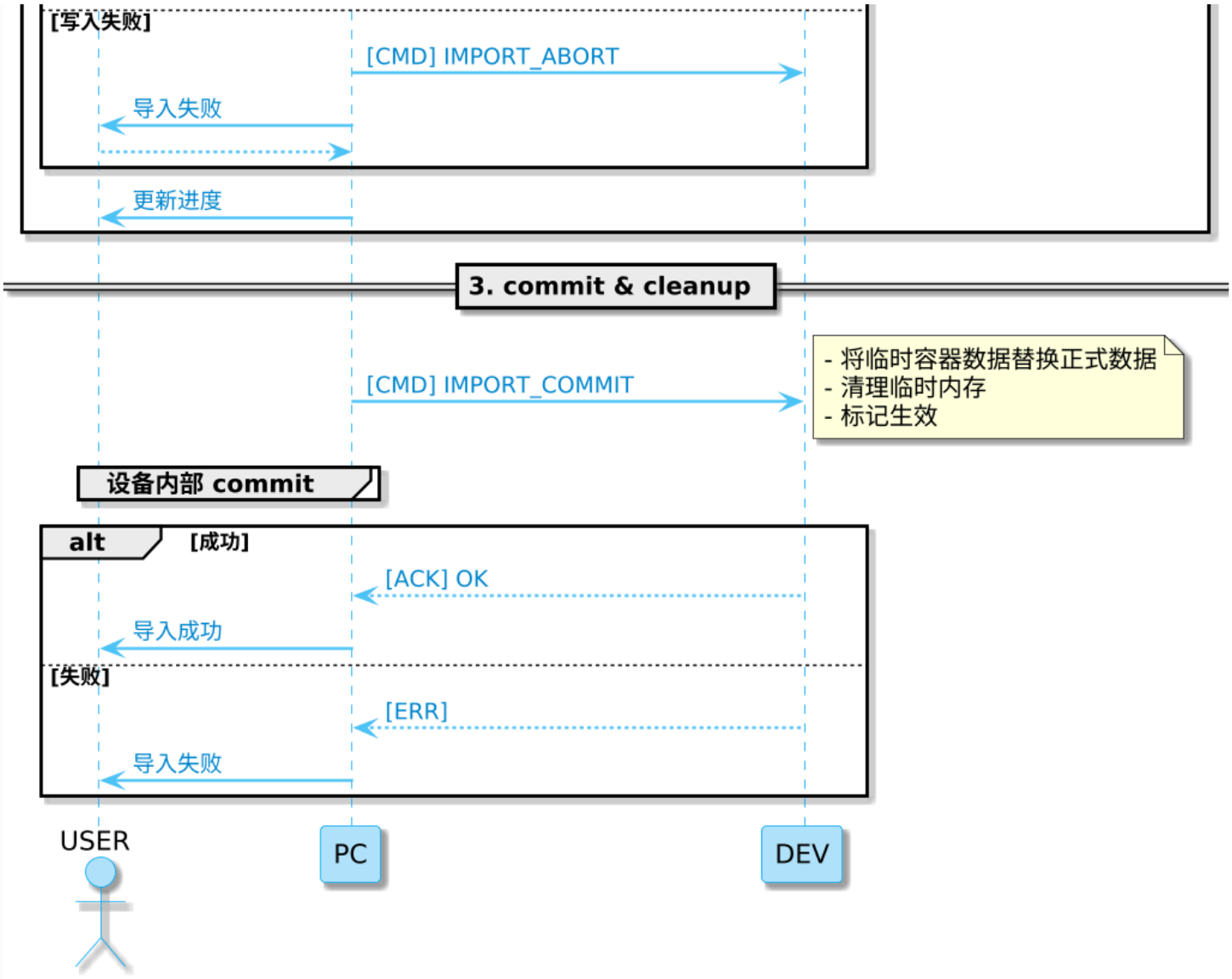
导出时序


USER
PC
DEV
1. task init
启动备份 (统一/单个)
[CMD] QUERY_META
[ACK] META_INFO(CONFIG,AppList,ImgList)
PC 初始化导出文件：
- 写 FileHeader(占位)
- 初始化 Index 缓存
- Data区准备写入
2. data transfer
loop
[每个程序 (App_ID)]
程序基础包 (APP)
[CMD] GET_APP(App_ID)
alt
[成功]
[DATA] payload(App定义协议原始数据)
写入 Data区：
[UnitHeader + payload]
UnitHeader:
- type = APP
- size = payload长度
- id = 自增ID
- checksum
payload:
- 原始协议数据（不解析）
记录 IndexEntry(type=APP, offset, size, id)
[失败]
退出流程（错误提示）

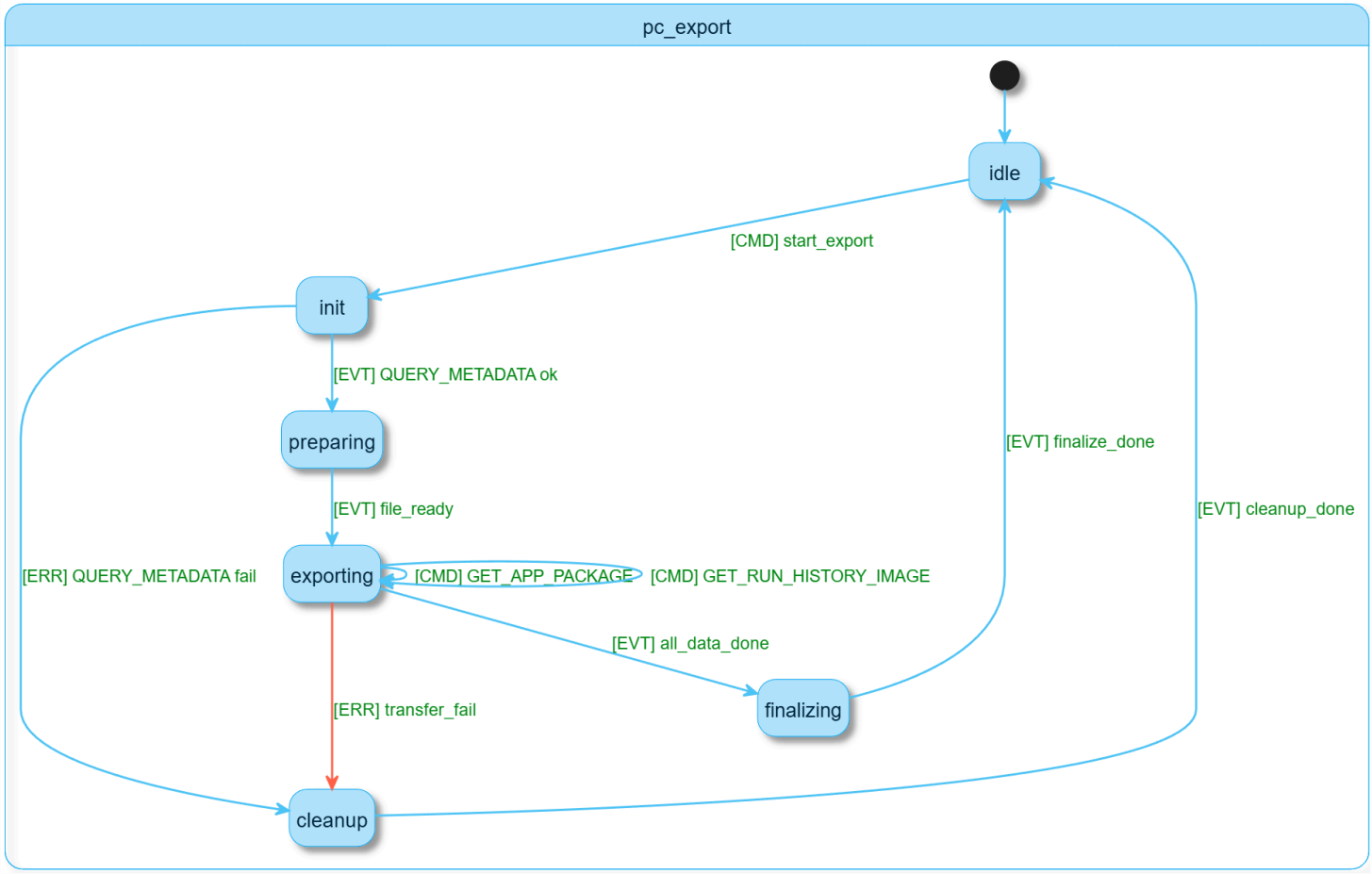

opt
[运行图像历史 (IMAGE)]
loop
[每张图像]
[CMD] GET_IMAGE(App_ID, Index)
alt
[成功]
[DATA] payload(图像协议原始数据)
写入 Data区：
[UnitHeader + payload]
UnitHeader:
- type = IMAGE
- size
- id
- checksum
记录 IndexEntry(type=IMAGE, offset, size, id)
[失败]
退出流程
更新进度
3. commit & cleanup
写入 Index区：
IndexEntry:
- type
- offset（指向UnitHeader）
- size
- id
回填 FileHeader：
- index_offset
- index_size
导出完成（.bak）
USER
PC
DEV

导入时序

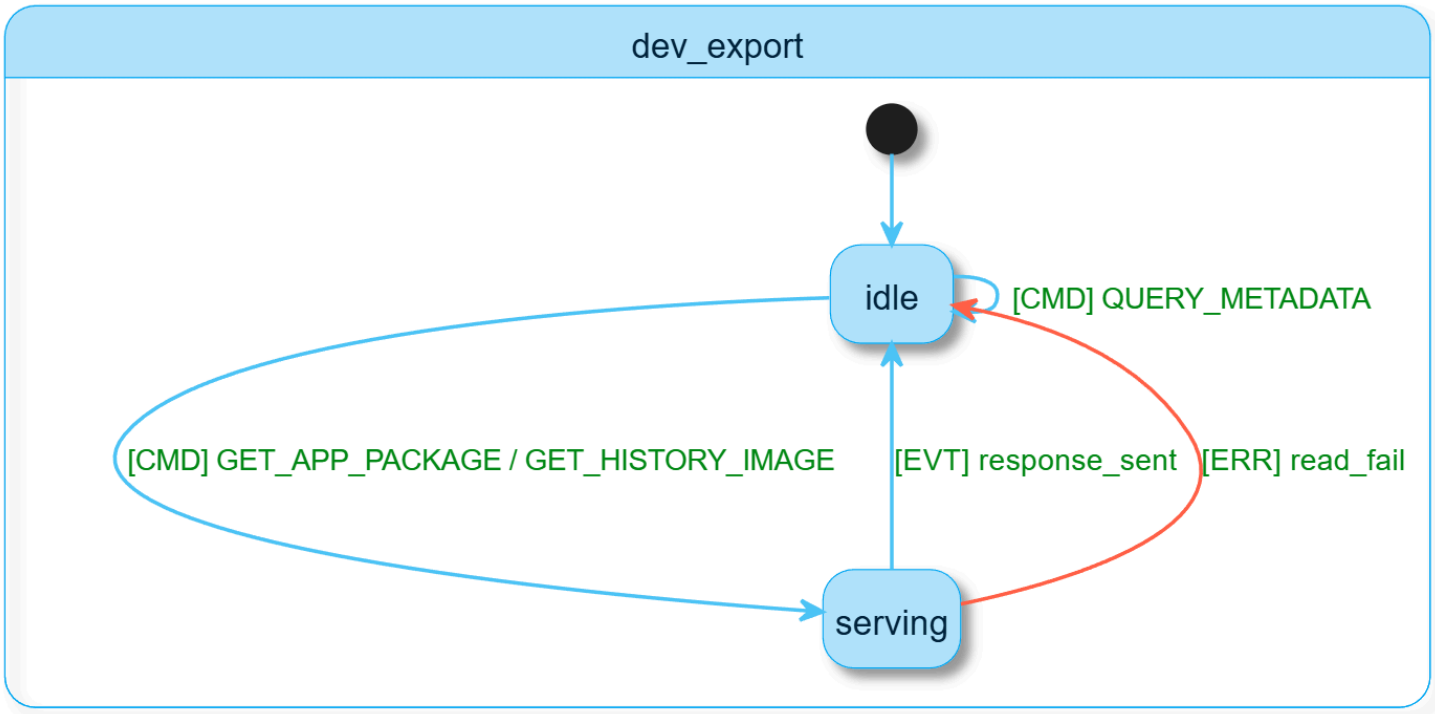

USER
PC
DEV
1. task init
选择备份文件 (.bak)
读取 FileHeader
alt
[文件合法]
定位 Index区 (index_offset)
读取 IndexTable
[文件非法]
提示错误
[CMD] IMPORT_BEGIN
[ACK] OK
设备进入导入态：
- 清空临时容器
- 初始化导入状态机
2. data replay
loop
[每条 IndexEntry]
根据 offset 读取 UnitHeader + payload
解析 UnitHeader（本地）：
- type
- size
- id
payload 不解析
alt
[type == APP]
[CMD] IMPORT_APP(payload)
[type == IMAGE]
[CMD] IMPORT_IMAGE(payload)
- 接收 payload
- 校验 checksum
- 解包协议
- 填充到临时内存结构：
• APP list
• IMAGE list
- 标记状态 OK/ERR
设备内部处理
alt
[写入成功]
[ACK] OK


[写入失败]
[CMD] IMPORT_ABORT
导入失败
更新进度
3. commit & cleanup
[CMD] IMPORT_COMMIT
- 将临时容器数据替换正式数据
- 清理临时内存
- 标记生效
设备内部 commit
alt
[成功]
[ACK] OK
导入成功
[失败]
[ERR]
导入失败
USER
PC
DEV

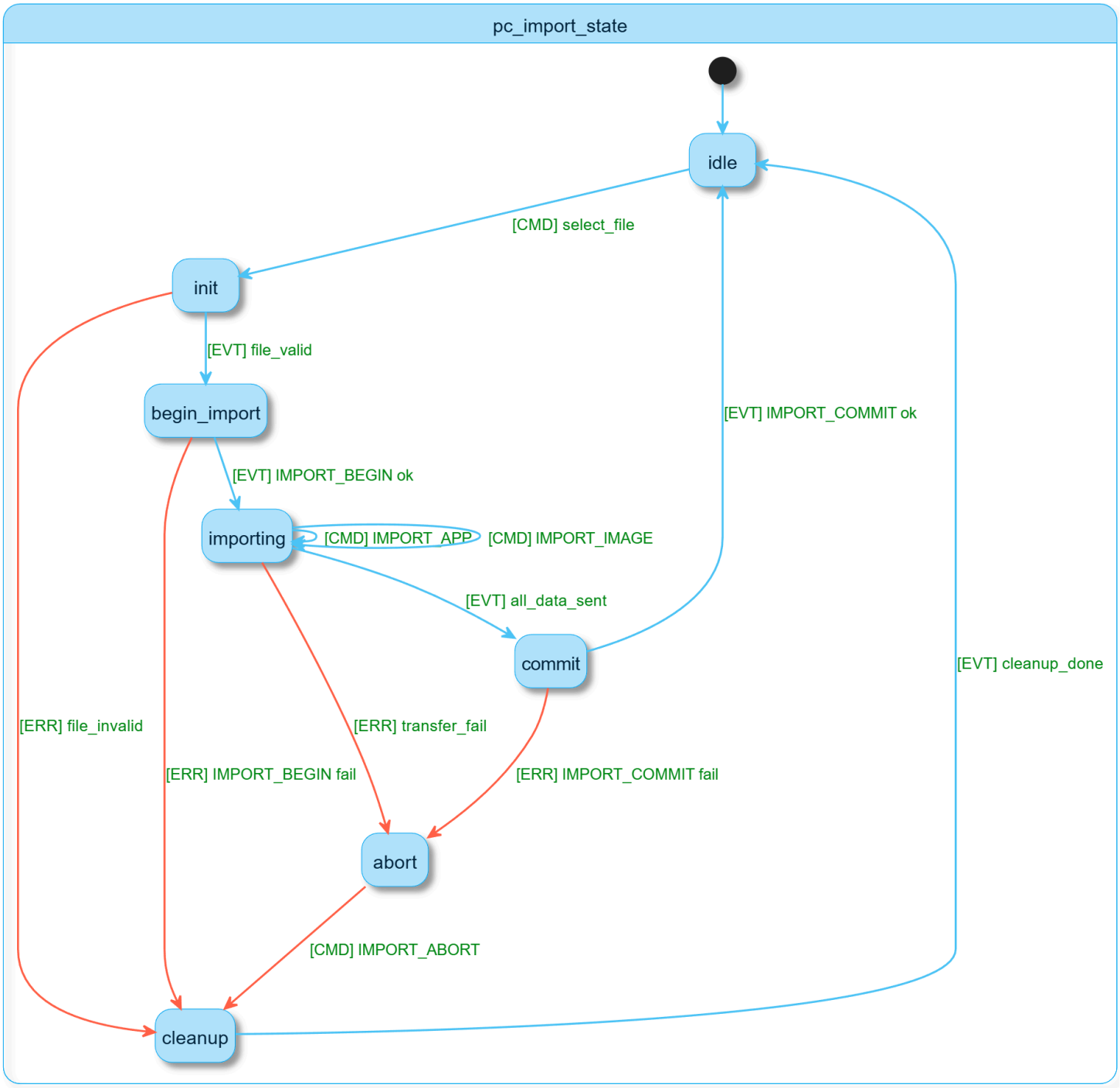
导出时，上位机状态


pc_export
idle
[CMD] start_export
init
[EVT] QUERY_METADATA ok
preparing
[EVT] file_ready
exporting
[CMD] GET_APP_PACKAGE
[CMD] GET_RUN_HISTORY_IMAGE
[ERR] QUERY_METADATA fail
[EVT] all_data_done
finalizing
[EVT] finalize_done
[EVT] cleanup_done
[ERR] transfer_fail
cleanup

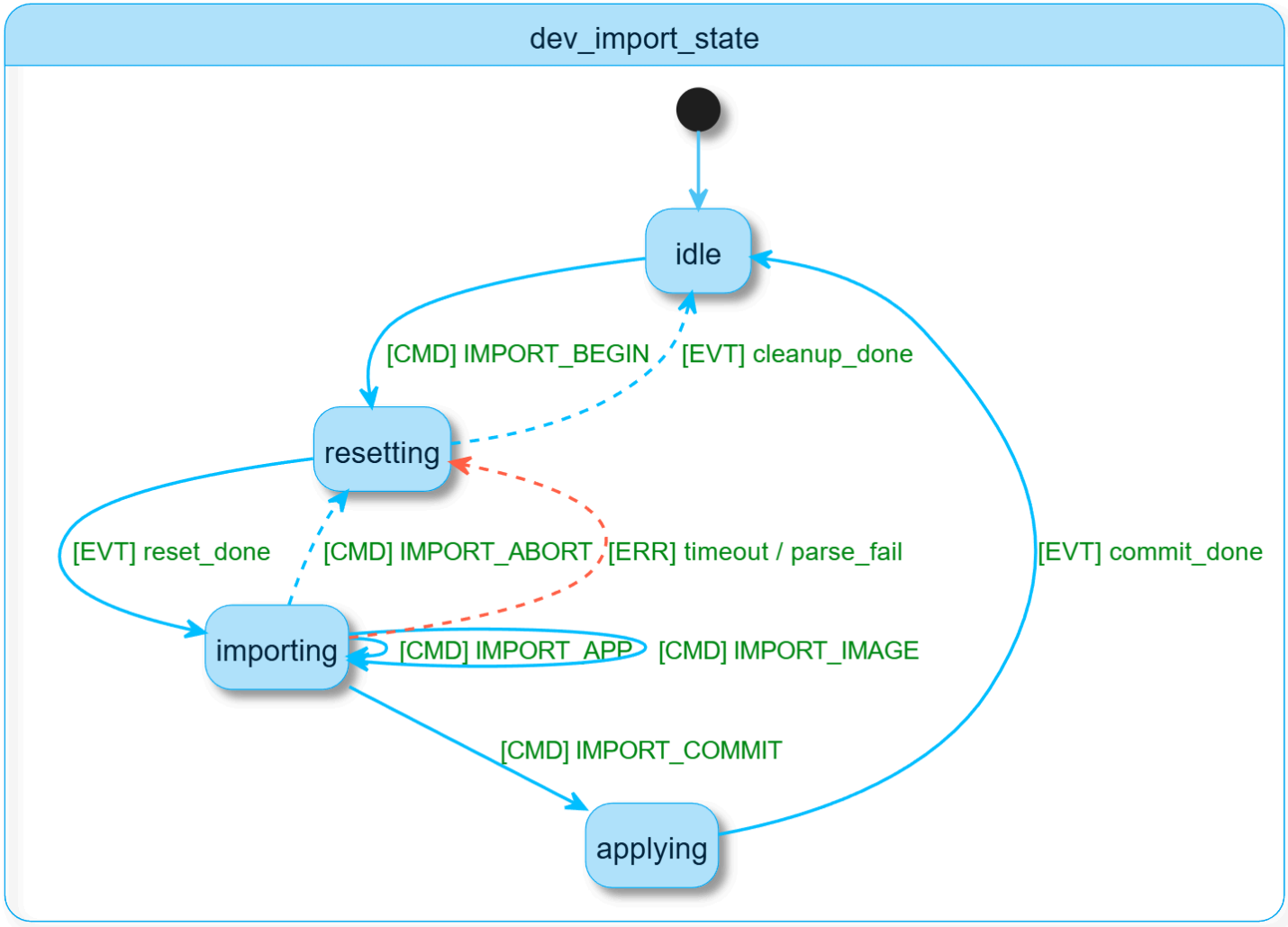
导出时，下位机状态


dev_export
idle
[CMD] QUERY_METADATA
[CMD] GET_APP_PACKAGE / GET_HISTORY_IMAGE
[EVT] response_sent
[ERR] read_fail
serving

导入时，上位机状态


pc_import_state
idle
[CMD] select_file
init
[EVT] file_valid
begin_import
[EVT] IMPORT_BEGIN ok
importing
[CMD] IMPORT_APP
[CMD] IMPORT_IMAGE
[EVT] all_data_sent
commit
[EVT] IMPORT_COMMIT ok
[EVT] cleanup_done
[ERR] file_invalid
[ERR] transfer_fail
[ERR] IMPORT_BEGIN fail
[ERR] IMPORT_COMMIT fail
abort
[CMD] IMPORT_ABORT
cleanup

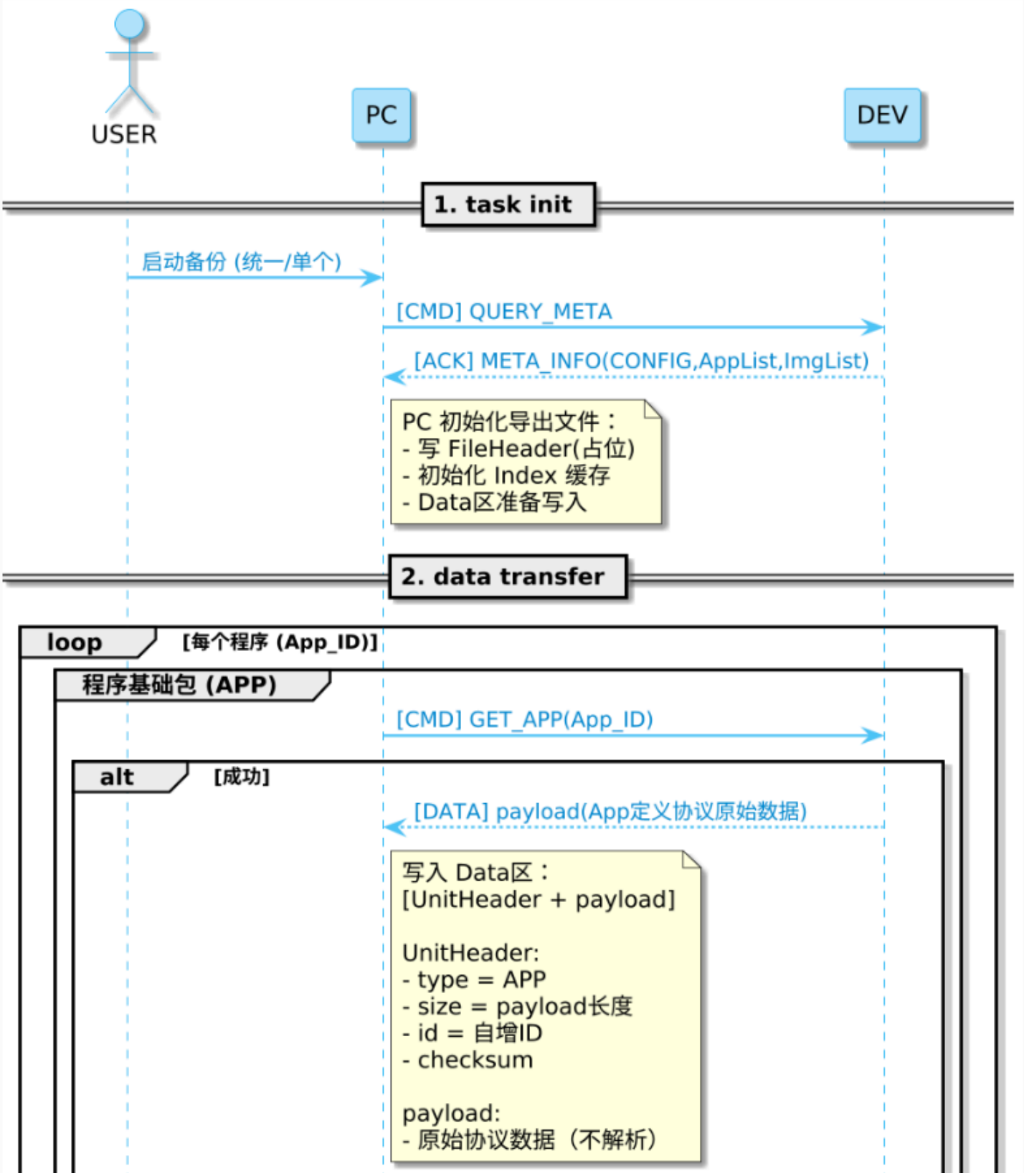
导入时，下位机状态

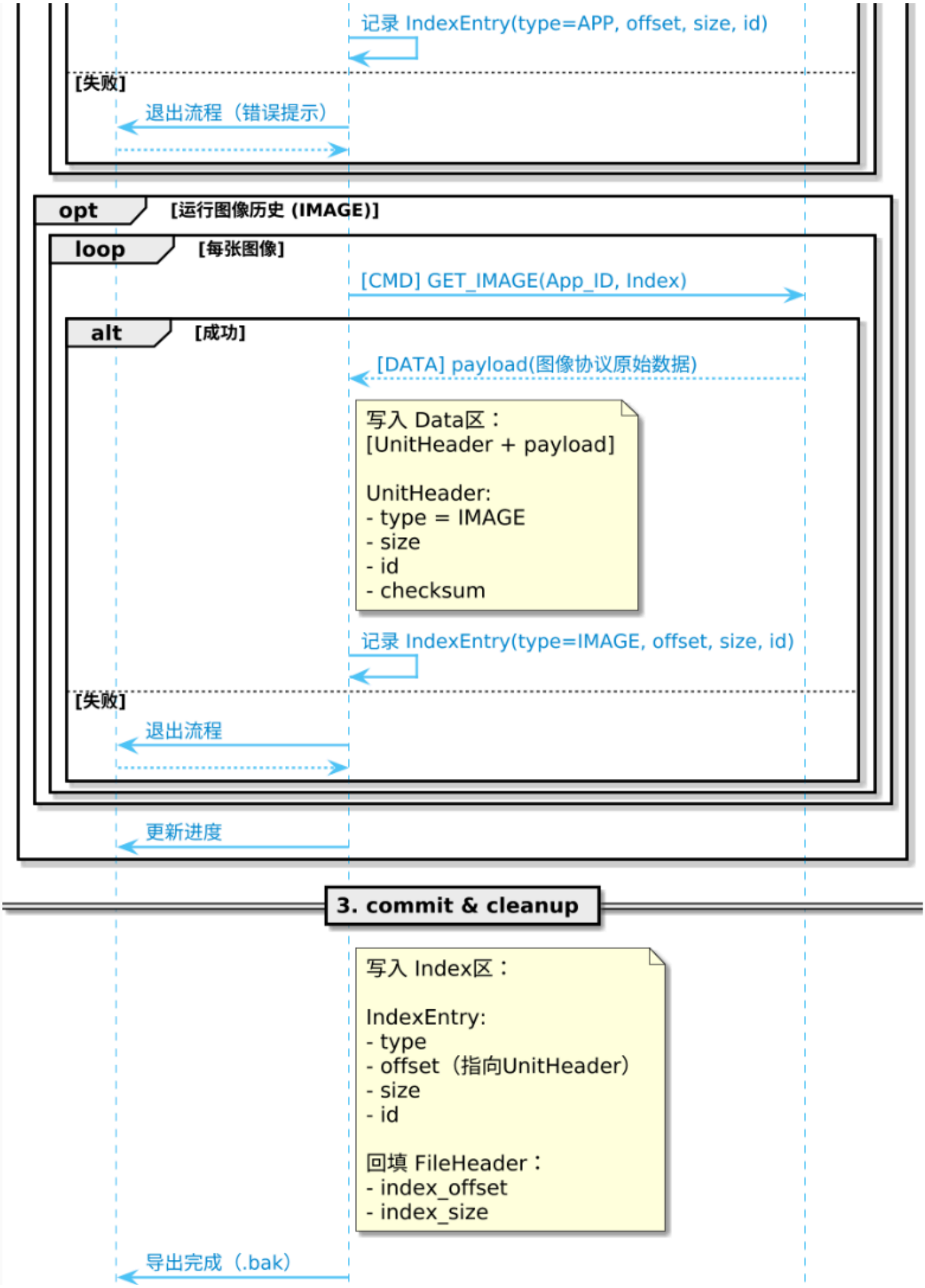



```
# 简概
+-------------------+
| File Header       |  <- 固定长度
|-------------------|
| Data Unit 1       |  <- UnitHeader + payload (主控/原图/缩略图等)
| Data Unit 2       |
| Data Unit 3       |
| ...               |
|-------------------|
| Index Table       |  <- 支持随机访问
+-------------------+
```

时序图

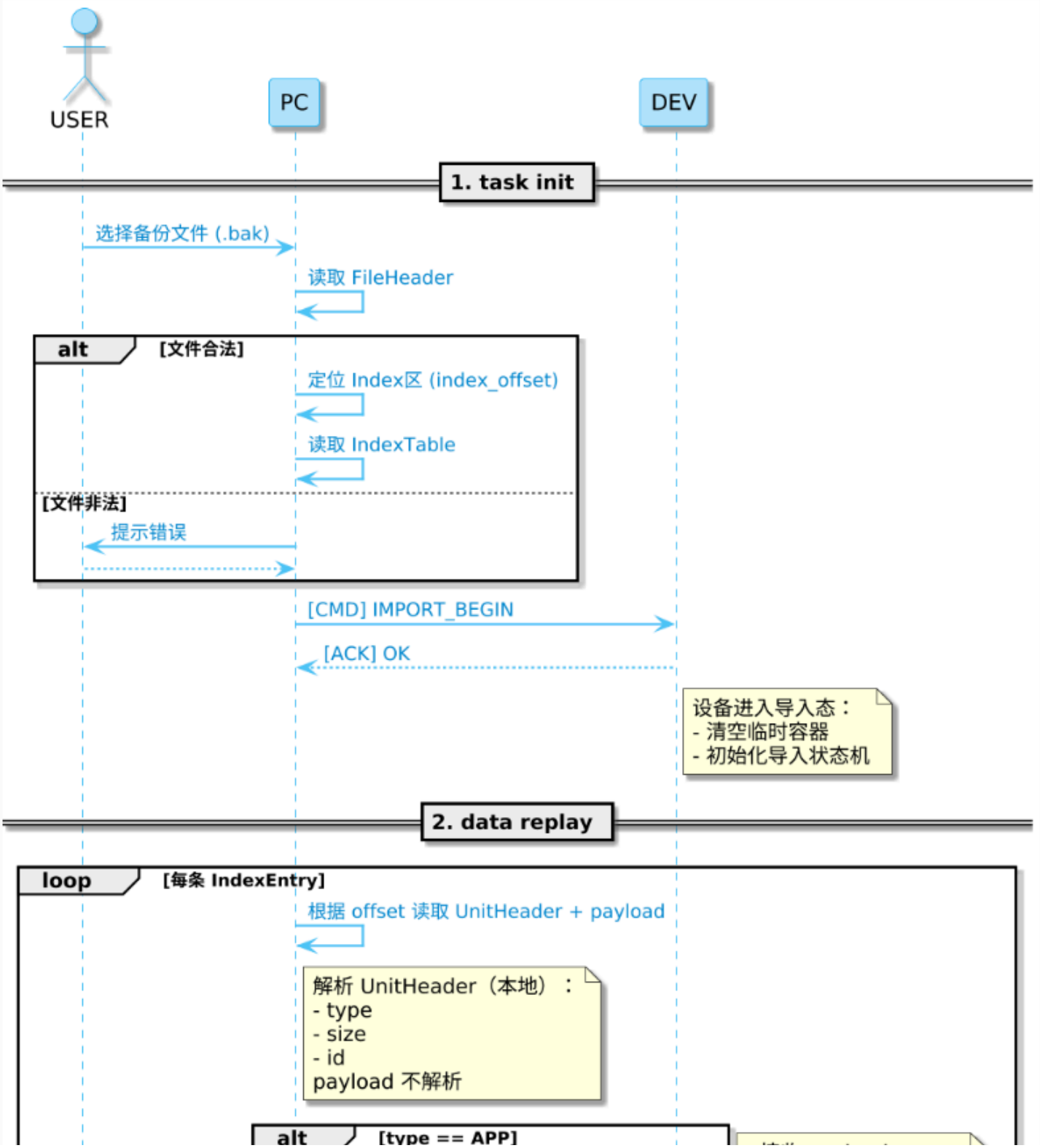
导出时序


USER
PC
DEV
1. task init
启动备份 (统一/单个)
[CMD] QUERY_META
[ACK] META_INFO(CONFIG,AppList,ImgList)
PC 初始化导出文件：
- 写 FileHeader(占位)
- 初始化 Index 缓存
- Data区准备写入
2. data transfer
loop
[每个程序 (App_ID)]
程序基础包 (APP)
[CMD] GET_APP(App_ID)
alt
[成功]
[DATA] payload(App定义协议原始数据)
写入 Data区：
[UnitHeader + payload]
UnitHeader:
- type = APP
- size = payload长度
- id = 自增ID
- checksum
payload:
- 原始协议数据（不解析）


记录 IndexEntry(type=APP, offset, size, id)
[失败]
退出流程（错误提示）
opt
[运行图像历史 (IMAGE)]
loop
[每张图像]
[CMD] GET_IMAGE(App_ID, Index)
alt
[成功]
[DATA] payload(图像协议原始数据)
写入 Data区：
[UnitHeader + payload]
UnitHeader:
- type = IMAGE
- size
- id
- checksum
记录 IndexEntry(type=IMAGE, offset, size, id)
[失败]
退出流程
更新进度
3. commit & cleanup
写入 Index区：
IndexEntry:
- type
- offset（指向UnitHeader）
- size
- id
回填 FileHeader：
- index_offset
- index_size
导出完成（.bak）

USER
PC
DEV

导入时序


USER
PC
DEV
1. task init
选择备份文件 (.bak)
读取 FileHeader
alt
[文件合法]
定位 Index区 (index_offset)
读取 IndexTable
[文件非法]
提示错误
[CMD] IMPORT_BEGIN
[ACK] OK
设备进入导入态：
- 清空临时容器
- 初始化导入状态机
2. data replay
loop
[每条 IndexEntry]
根据 offset 读取 UnitHeader + payload
解析 UnitHeader（本地）：
- type
- size
- id
payload 不解析
alt
[type == APP]

[CMD] IMPORT_APP(payload)

[type == IMAGE]

[CMD] IMPORT_IMAGE(payload)

- 接收 payload
- 校验 checksum
- 解包协议
- 填充到临时内存结构：
  - APP list
  - IMAGE list
- 标记状态 OK/ERR

设备内部处理

alt [写入成功]

[ACK] OK

[写入失败]

[CMD] IMPORT_ABORT

导入失败

更新进度

3. commit & cleanup

[CMD] IMPORT_COMMIT

- 将临时容器数据替换正式数据
- 清理临时内存
- 标记生效

设备内部 commit

alt [成功]

[ACK] OK

导入成功

[失败]

[ERR]

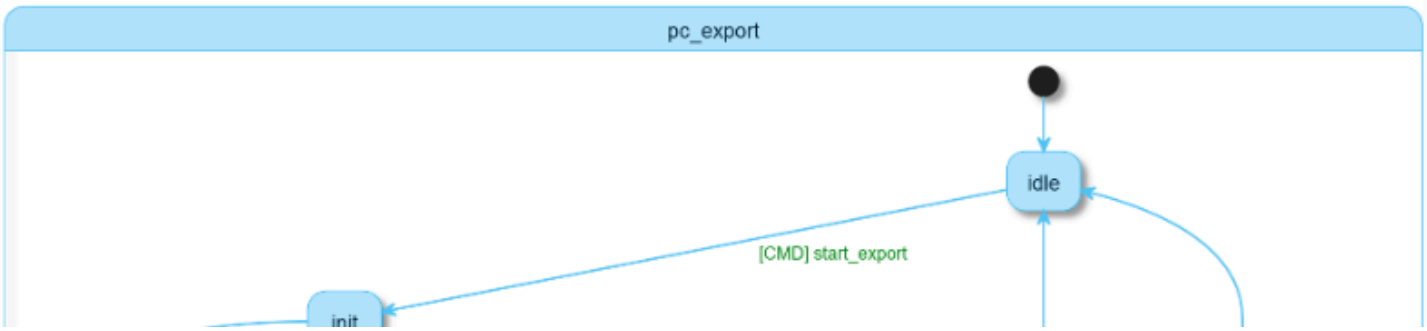
导入失败

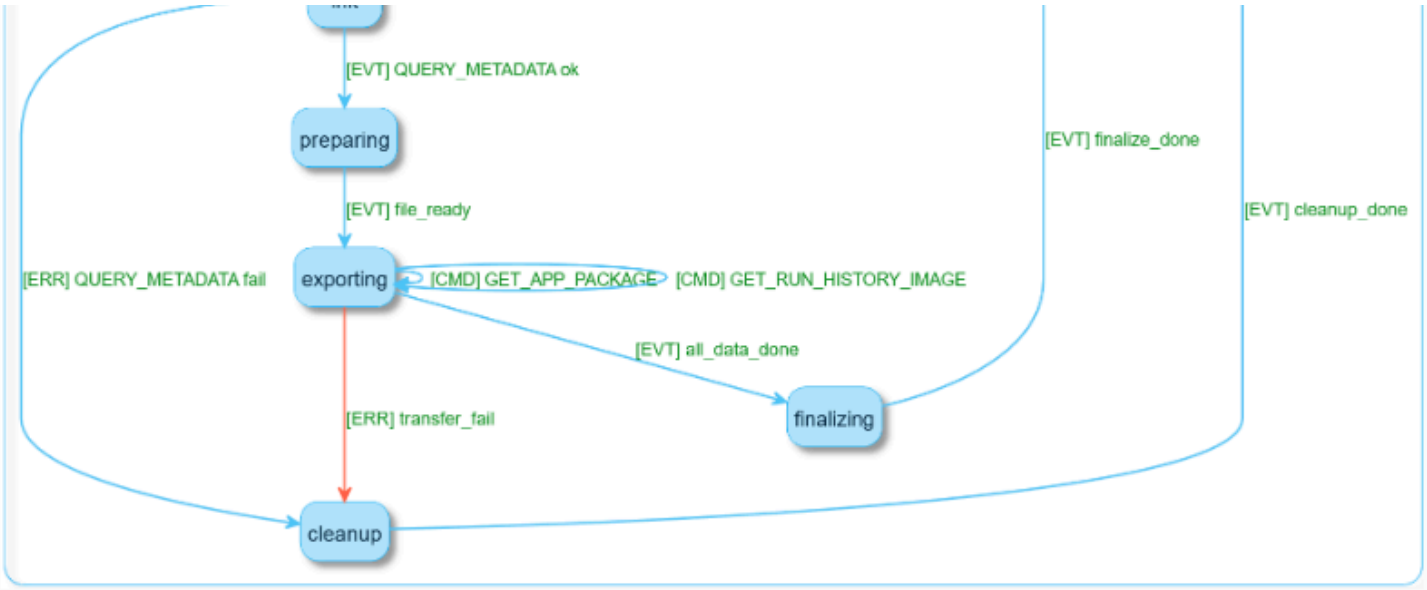
USER

PC

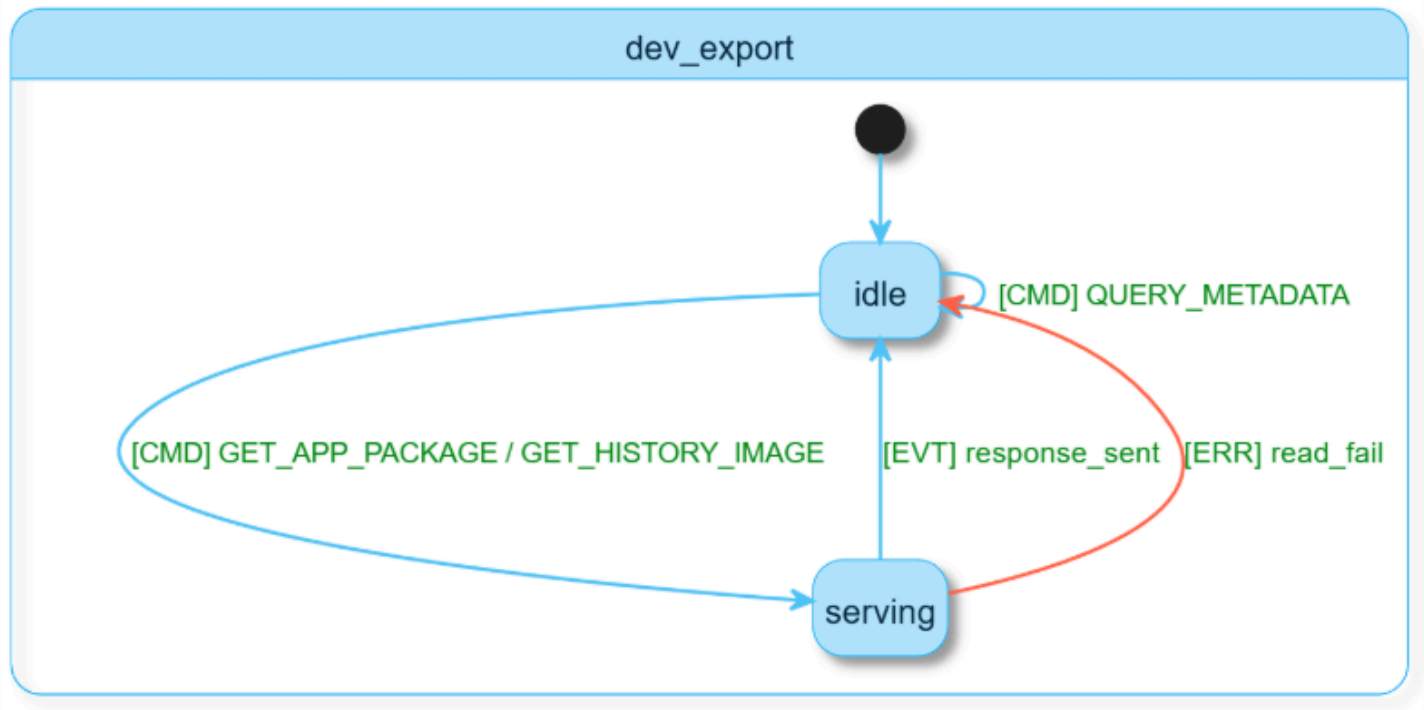
DEV

导出时，上位机状态

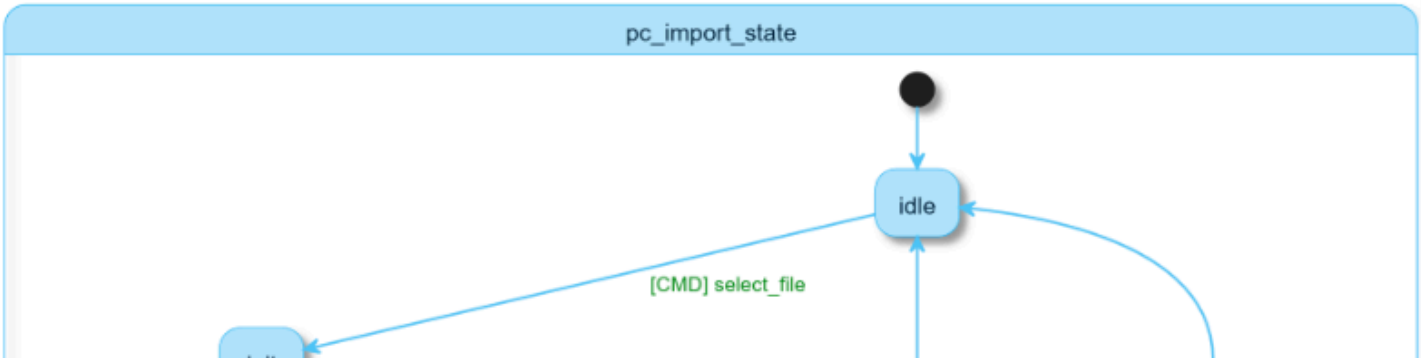


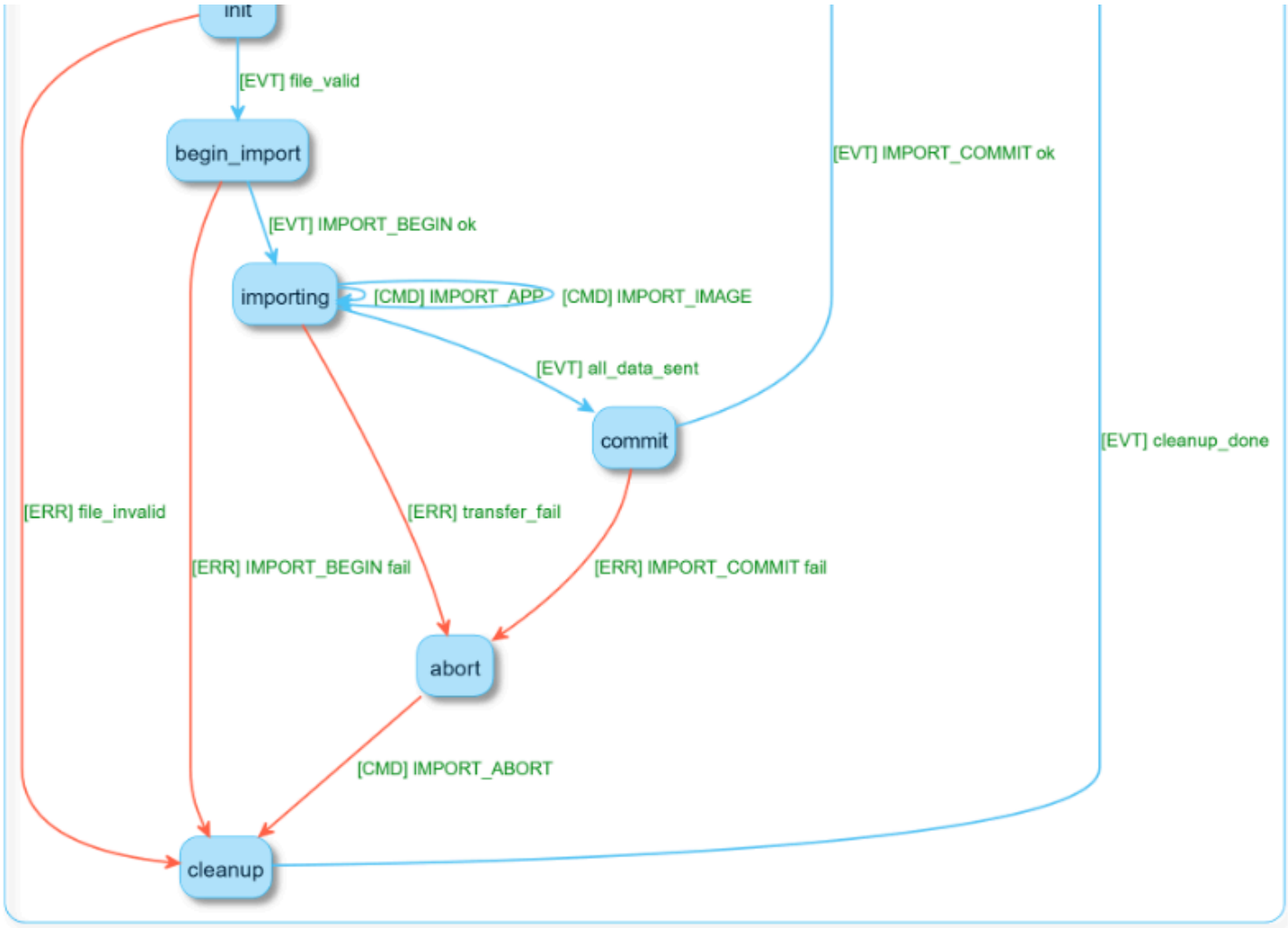


导出时，下位机状态

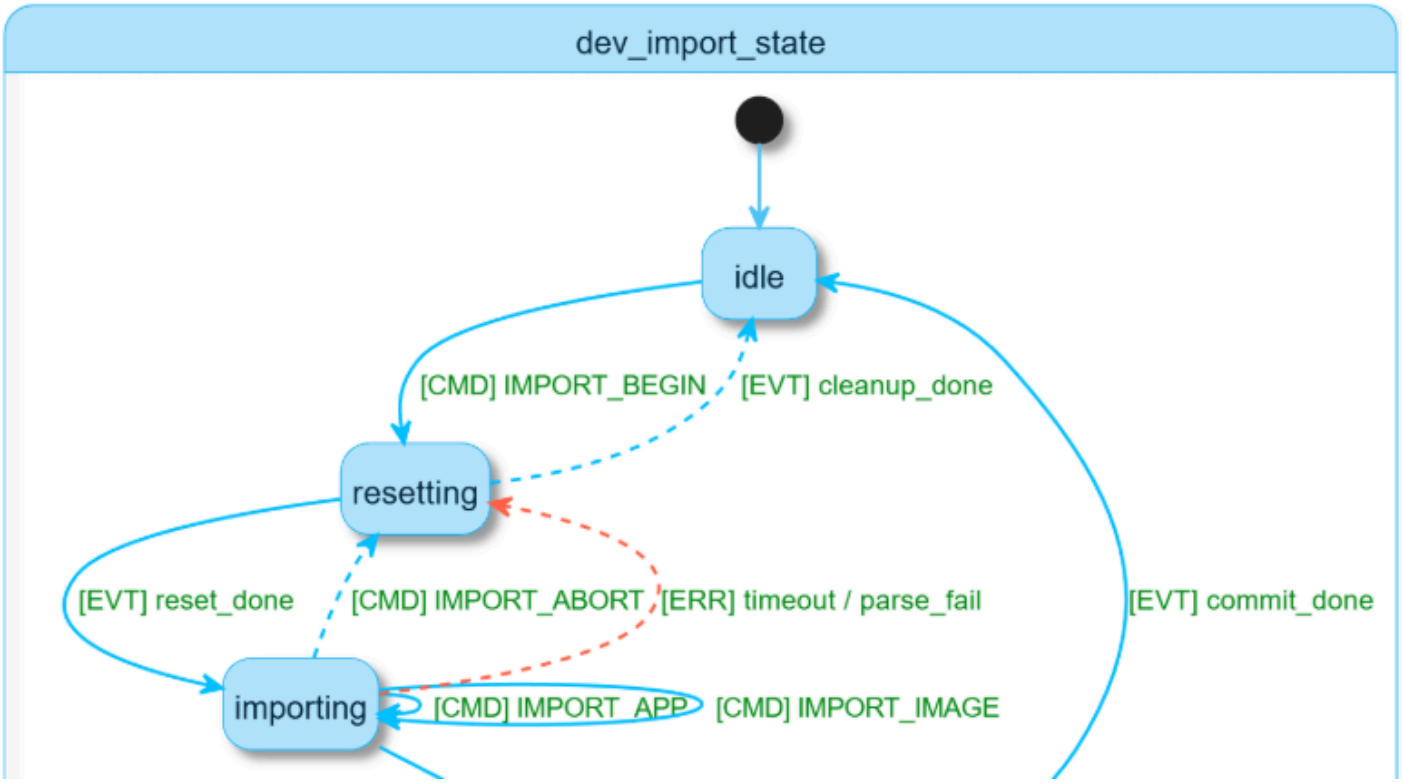


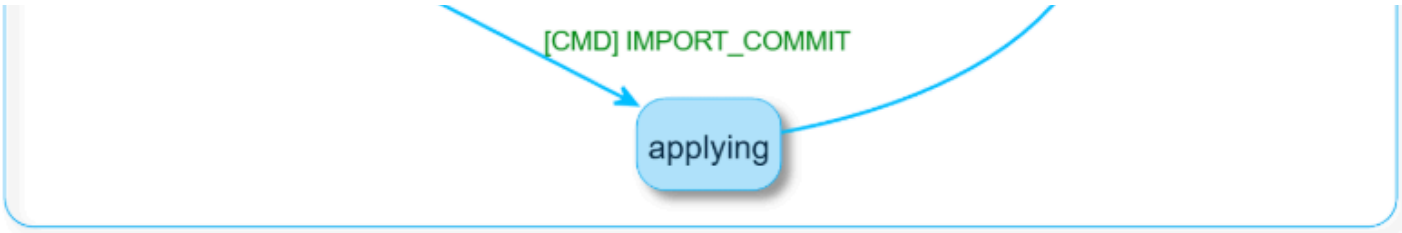
导入时，上位机状态




init
[EVT] file_valid
begin_import
[EVT] IMPORT_BEGIN ok
importing
[CMD] IMPORT_APP
[CMD] IMPORT_IMAGE
[EVT] all_data_sent
commit
[EVT] IMPORT_COMMIT ok
[EVT] cleanup_done
[ERR] file_invalid
[ERR] transfer_fail
[ERR] IMPORT_BEGIN fail
[ERR] IMPORT_COMMIT fail
abort
[CMD] IMPORT_ABORT
cleanup

导入时，下位机状态


dev_import_state
idle
[CMD] IMPORT_BEGIN
[EVT] cleanup_done
resetting
[EVT] reset_done
[CMD] IMPORT_ABORT
[ERR] timeout / parse_fail
[EVT] commit_done
importing
[CMD] IMPORT_APP
[CMD] IMPORT_IMAGE




```
# 简概
┼──────────────────┼
│ File Header       │  <- 固定长度
├──────────────────┤
│ Data Unit 1       │  <- UnitHeader + payload (主控/原图/缩略图等)
│ Data Unit 2       │
│ Data Unit 3       │
│ ...               │
├──────────────────┤
│ Index Table       │  <- 支持随机访问
┼──────────────────┼
```